ME 5405 Machine Vision

CA – AY25/26 Semester 2

Computing Project

You will be assigned to a random group of 3 students to work on the computing project. The software must be developed using MATLAB. You are encouraged to rely on your own implementations for the different steps of this project. You will be rewarded doing so. Otherwise, you are allowed to use MATLAB's Imaging Toolbox.

Images 1 and 2 are 64x64, 32 level images. These images are shown as coded arrays that contain an alphanumeric character for each pixel. The range of these characters is 0-9 and A-V, which corresponds to 32 levels of gray.

In your report, you should include the followings for Image 1 and Image 2.
1. an introduction to the problem,
2. a description of your algorithm and flow chart,
3. screen captures of every stage of the image processing, and
4. a discussion and conclusion including an explanation on why you choose the method employed in your project, which investigation you performed, and what lessons you learned.

You should upload your report and software to CANVAS-ME5405-Files-Student Submission by **19 April 2026 23:59 (Sunday)**.

This is a group project. Please submit only one set of report and codes. All members of the group will receive the **same score** unless a clarification email is received from the group. Name your submission as *Group XX.zip*. Please indicate the names and IDs of all group members on the cover page of your report.

Image 1: Chromosomes (Available on CANVAS-ME5405-Files-Group Project – chromo.txt)

Perform the following tasks:
1. Display the original image on screen.
2. Threshold the image and convert it into binary image.
3. Determine a one-pixel thin image of the objects.
4. Determine the outline(s).
5. Label the different objects.

Image 1

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EJNPPRSTSTSTTTURQSVVVVUUUVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVVV
CFJMNNNNNOPORPQOPORRSQQRQQTSRTSTTTUUTURTUVVVVVVVVVVVUVVVVVVVVVVVVV
BEILKMMLMLONONPNNNNPOOOPOPTURSQRRRSUTTRSSSSTTSUTVUVUUVVVVVVVVVVVVV
BFHKLMOMNONNOOOPNMNNOPPOMPORRQQQPRSTSRPSQPTSQTSRTSSUUVUUVVVVVVVVV
AEHKLLMNNMMMOPPONNNOOONPONQPQPPQQRRTSTRQRSTSSSUTTTVVUVVVVVVVVVVVVV
BEGILLMNNMMMPOPONNNNOPPOMOPQRQQPRRSRSRSQPRTTTSSUUSTUVVVUVVVVVVVVV
BDHJLMMNOMNOMOOOOLNMOPOONOQORRQPPQSTURQQRQSSTSTTTTTVUUVVVVVVVVVV
BEHKKMNOONNMOMNNMLMLMNQNNOOQQQROPPSTRSRQQQSQTTTTTTTSUVVVVVVVVVVU
CFHJKKMNNNMNLLHKMNLGAPONNNPQQPPOOPQRSQQRQRQTTVTTTRTSUVVVVVVVVVQH
DFIKLMMMONMMIE9DLLLE6ALNOMPQRQPPOPQSRRQOQPRTTUTSURSTUTUVVVVVVSNK
DFIKKMMMMMMLF86CLNLC69KNNMPRRPPOOPSTRSQRPRRSUSSRSRSTUUUVVVVVVUUU
CGJKKLMNMMLKE46FMMLF67JNLOOPRRPPOQRSRSRPQRTTTTTTTTSUUUVVVVVVVVVV
DFHKKLMMNNLID67HOMMI87EOONQPSRPOPQQUURRPORRTTTRRTTTUTUVVVVVVVVVV
DFIJKMNNNMLJC5ALOMNKC9GNPPPRRRRPQQRTTRRQQSSSTSSQRTTSVUVVVVVVVVVV
DFIKMMOOOOMKE49KONNK97HPQPQTUSQQPRSTSRRRRSUUUTRSSSUTVUVVVVVVVVVV
DGJLMMNNOOMKE69JNNMI63GPOPRRTTRQQSTUTSRQRRTTUVTTTTUTVVVVVVVVVVVV
EGJLMNNOONOLD45HMOMJA6FQRQRSUTTRRRVTSTRRRTTUTTTSSTUTVVVVVVVVVVVV
DGJLLLOOONMNH64BLMMLD9IPQQRSTTRRRTVTTRSTTSUVUTSTTRUVVVVVVVVVVVVV
EHKLNMNONOMMLF69KMLKE7HQQQSSUUSSTUVVTSTRTTVVVUTTUUTUVVVVVRVVVVVV
EHJLLOOOPPOONJCAINNKG6CPPQRSUTRSUVVUSRTSVVUVTRRTTTUVVVVVVVVVVVV
EIJLMOPPOOOMOMJDGMMGA8FNQRSVUTTRSUUUTTTTUVVVUUTUTTTVUVVVVVVVVVVV
EHKMMNOPPONMNMHADMME47HMOOSRUTUTSVVVUTSTUVVVUURTTUVVVVVVVVVVVVVV
EHLMNOOOPPONNOH79JMG66IPNPRTVURSTUVURTSTUVVVVURSUUVVVVVVVVVVVVVV
FHKLMNOPPNNOPPK65GME6EOQOPSTTRQQTVVVTTRSVVVUVTRRTUTVVVVVVVVVVVVV
FHKMNOPOPPONNOOE6EMD9KPROOTTUSQQSVVUTRRTVVVVTSRRRSUVVUUUVVVVVVVV
EHKMNNPPOOOOMOOL85GGENQQPQTTSSQQTVVUSRQSVVUUSSRRTTTUVVVVVVUVVVVV
EHKMNPOPOQPOPPPOI9BEGPRQQRUTTSRQTUVTQRRTTVTURRSRRUTVVUUTUVUVVVVV
FIKMNPQQQPOOOPQNNGBBDLPRRTTTTRRTVVVTRRRUVVTQRRRRTTTVVRORTVVVVVVV
GILNOQRQQPPPQPPPLE8FF9CKQTVUSRRUVVVSQRRUVVURRRTSVSVTRF7ITTUVVVVV
GJLOPQQRPPPPQQQNJ84FK74BOUVUURTVVVVTRQUVVUTSRRTUQLMSQA2ERTUVVVVV
FJLNPPPPQPPQRQPLA11EJA25LUUUUTVVVVVUQSVVVUVRSTUTP56NRD2DQUUVVVVV
GHLOPRPPQPQQRPOH717LNE309PVSSSVVVVVSUUVVVUTTTUUUO32LSF29QVUVVVVV
FJMPPQQPPPQQRQPF47JQQMA25PTVSSVVVVVVUVVVVTSUVTVVQD7IQD5EQVVVVVVV
GJLOPQQPPQQRQRPE26LTQNE59PRSRSVVVVUTVVVVVVSVUVVVTJ8BMB9NTTTVVVVV
GJNQPRRRQRRRRPM927OTQMF7BQRQRTVVVVSTVVVVVVUVVVVVVN68E6BNLHHLQLLH
GJPQRRRRQSRSRQH52DRRQLEBKRSRSVVUTSTTVVVVVVVVVVVVVTMFB5BE739EF857
HKOQRPQRSSSRQPI43GSQQH69NSSSUUVVTTTUVVVVUSVVVVVVVVRMGHHHIHOQLD8G
HKOQQQSSTSTRRQOHEMSQQNHKTUTSVVVRVTUVVVVTTUUVVVUVVTQEERUUVVVVVTTV
ILPQRRSTTTTSRRRRSTRRTTUUUUVVVUTTVVVVVSSUUVVVVTQRJ9ERVVVVVVVVVV
IMPPRRSTTTTTTTTSTSTTVVVUTUVVVVUUVVVUTRTVVVUTSRK55LVUVVVVVVVVV
IMPRTSSUTTTSTTUTTSSTTVVTTVVVVVVVVVVVVURUVVVUTRRRK47QVVUVVVVVVVV
JNPQTTUUUTTTTVTVUTUUVVUTTUVVVVVVVVVVVVVVVVVVVTSTTK8CRVVUTVVVVVVV
KOPRTTUVUUTTTVVUUTTVVVUTUVVVVUVVVVVVVVVVVVVVVTUVUO57NVSTTUVVVVVV
KNQTUUVVVUVVVVUTTUVVUTTUUVVVVVVVVVVVVVVVVVVV
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Image 2: Characters (CANVAS-ME5405-Files-Group Project - charact1.txt)

Perform the following tasks:

1. Display the original image on screen.
2. Create a binary image using thresholding.
3. Determine a one-pixel thin image of the characters.
4. Determine the outline(s) of characters of the image.
5. Segment the image to separate and label the different characters.
6. Arrange the characters in one line with the sequence: **AB123C**
7. Rotate the output image from Step 6 about its center by 30 degrees.
8. Using the training dataset provided on CANVAS (*p_dataset_26.zip*), train the (conventional) classification method of your choice (i.e., self-ordered maps (SOM), k-nearest neighbors (kNN), or support vector machine (SVM)) to recognize the 6 characters ("1", "2", "3", "A", "B", or "C"). You should use 75% of the dataset to train your classifier, and the remaining 25% for validation (testing). Then, test your trained classifier on each character in Image 2, reporting the final classification results. **Do not use the characters in Image 2 as training data for your classifier.**
9. Throughout step 8 (training of the classifier), also experiment with pre-processing of the data (e.g., padding/resizing input images) as well as with hyperparameter tuning. In your report, discuss how sensitive your approach is to these changes.

Image 2

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0000000004MJ0000000000000000000002STRRD0000000000000000000NNLM9000000
000000004LMH00000000000000000005OMPPOMO0000000000000008JKHOJJD00000
000000BONMMI0000000000000000000KOE00DNM80000000000000IJ6003CI10000
0000007FOKMH0000000000000000000NL0000HMD00000000000004KE0000BJ50000
0000000009MI00000000000000000BA0000FME0000000000000000000EJ30000
0000000008MH000000000000000000000000KNA000000000000000000FJ900000
0000000008MI0000000000000000000000JMK100000000000000000007JJH000000
0000000008MI00000000000000000000000LNL4000000000000000000004CIJ40000
0000000008MH000000000000000000000MOI3000000000000000000000007OF0000
0000000006MH00000000000000000000LOD0000000000000000005000001OH0000
0000000007MH000000000000000000000JMB00000000000000000000DJB00005HO0000
0000000007MH000000000000000000000AMMJADFID0000000000000003IK4003OID0000
00000000061D000000000000000000000JPOOOOONK000000000000000AKJJIIJF10000
0000000000100000000000000000000000BAAAABA74000000000000000005EHOE7000000
0000000000000000000000000000000000000000000000000000000000000000000
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